

# SIADS 696 Project Proposal: Baseball Pitch Identification and Clustering Model

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## Project Overview

This project uses MLB Statcast pitch-level data from the `pybaseball` library to study baseball pitch classification and pitch archetypes through supervised and unsupervised learning. Our goals are to (1) predict official MLB pitch labels using ball-flight telemetry and (2) discover finer-grained pitch subtypes that may provide additional scouting and pitch-design insight beyond standard categories such as fastballs, sliders, and changeups.

Key challenges include noisy Statcast spin-rate measurements, class imbalance across pitch types, and the difficulty of identifying meaningful unsupervised structure in high-dimensional pitch data.

## Related Work

Our closest reference is Statcast's automated pitch classification system and open-source pitch classification notebooks using `scikit-learn` and Baseball Savant data. Unlike prior work, we will compare multiple supervised models (Logistic Regression, Random Forest, LightGBM, Neural Networks) across different feature sets and extend the analysis with unsupervised clustering to identify 15–20 refined pitch subtypes evaluated against effectiveness metrics such as whiff rate and wOBA allowed.

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## Part A: Supervised Learning

### Dataset

We will use MLB Statcast pitch-level data accessed through the `pybaseball` Python package:  
[https://baseballsavant.mlb.com/statcast\\_search](https://baseballsavant.mlb.com/statcast_search)

The dataset contains ~700k–800k pitches per season with features including velocity, pitch trajectory (acceleration and velocity coefficients), spin rate, pitch start and end points, and official pitch labels.

### Learning Approaches

We will model pitch classification as a multiclass classification problem using: (1) Logistic Regression (baseline) (2) Random Forest (3) LightGBM (4) Multi-Layer Perceptron (MLP)

We will compare movement-only, velocity-plus-movement, and full Statcast feature sets to evaluate feature importance and model performance.

### External Tools

We plan to use: (1) `pybaseball` (2) `scikit-learn` (3) LightGBM/XGBoost (4) `pandas/numpy` (5) `matplotlib/seaborn`

## Evaluation & Visualization

Evaluation metrics will include accuracy, macro F1-score, per-class precision/recall, and confusion matrices. Visualizations will include pitch movement scatterplots, confusion matrices, and feature importance/SHAP plots.

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## Part B: Unsupervised Learning

### Dataset & Goal

Using the same Statcast dataset and features (without labels during training), we aim to determine whether pitches naturally cluster into more granular subtypes than existing MLB pitch categories and whether those clusters correlate with effectiveness metrics.

### Data Preparation

We will standardize features, handle missing values, and apply PCA/UMAP for dimensionality reduction and visualization.

### Unsupervised Methods

We will experiment with: (1) K-Means (2) Gaussian Mixture Models (GMM) (3) DBSCAN

### Evaluation & Visualization

We will evaluate clustering quality using silhouette score, elbow method, adjusted Rand index, and cluster purity against official pitch labels. We will also compare clusters using whiff rate and wOBA allowed. Visualizations will include UMAP/PCA scatterplots and cluster feature heatmaps.

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### Team Planning

| Team Member Responsibilities                                                        | Timeline                                              |
|-------------------------------------------------------------------------------------|-------------------------------------------------------|
| Ian MacKinnon - Supervised modeling, feature ablation, evaluation                   | Weeks 1–2: Data acquisition, cleaning, EDA            |
| Sachin Murthy - Data pipeline, preprocessing, infrastructure                        | Weeks 3–4: Supervised modeling and evaluation         |
| Asad Khan - Unsupervised modeling, clustering, dimensionality reduction, evaluation | Weeks 5-6: Clustering and visualization               |
|                                                                                     | Weeks 7–8: Final tuning, analysis, and report writing |